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PUSH-PULL VALVE

Abstract

A valve includes a valve body that has a flow path that connects the inlet and the outlet. A seal is disposed in the flow path. The seal has a closed state so as to block flow through the flow path and an open state so as to permit flow through the flow path. An armature is disposed in the valve body and is moveable with respect to the seal. An electromagnet is configured to selectively receive an electric current in a first current direction or a second, opposite current direction. When the electric current is in the first current direction the electromagnet actuates the armature to move the seal from closed to open and when the electric current is in the second current direction the electromagnet magnetically actuates the armature to move the seal from the open to closed.

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Background/Summary

BACKGROUND

[0001] Valves are ubiquitous in systems that handle fluids. One example is in a motor to control fuel flow for combustion. Such valves may include a valve armature that carries a valve member that seals against a valve seat. A spring may bias the armature to a closed position in which the valve member is seated in the valve seat so as to block flow. A solenoid coil serves as an actuator. When electric current is provided to the coil, a magnetic field is generated that causes the armature to move against the spring force and thereby lift the valve member off of the valve seat to permit flow. Once power to the solenoid is turned off, the spring moves the armature back to the closed position.

SUMMARY

[0002] A valve according to an example of the present disclosure includes a valve body that has a flow path, and a seal disposed in the flow path. The seal has a default closed state sealed against a seal seat so as to block flow through the flow path and an open state separated from the seal seat so as to permit flow through the flow path. An armature is disposed in the valve body and is moveable with respect to the seal. An electromagnet is configured to selectively receive an electric current in a first current direction or a second, opposite current direction. When the electric current is in the first current direction the electromagnet actuates the armature to move the seal from the closed state to the open state, and when the electric current is in the second current direction the electromagnet actuates the armature to move the seal from the open state to the closed state.

[0003] In a further embodiment of any of the foregoing embodiments, the electromagnet includes first and second coil windings that are oppositely wound.

[0004] In a further embodiment of any of the foregoing embodiments, the armature includes a rod portion, a yoke comprised of a permanent magnet and a keeper that surrounds the permanent magnet, and a shell that encases the yoke.

[0005] In a further embodiment of any of the foregoing embodiments, the rod portion has a tip that is in contact with the seal.

[0006] In a further embodiment of any of the foregoing embodiments, the shell is non-ferromagnetic and is formed of a titanium-based alloy.

[0007] In a further embodiment of any of the foregoing embodiments, the seal is an elastomer.

[0008] In a further embodiment of any of the foregoing embodiments, the shell is a material selected from the group consisting of titanium-based alloy, aluminum-based alloy, and polymer.

[0009] In a further embodiment of any of the foregoing embodiments, when the electric current is in the first current direction the electromagnet magnetically pushes the rod portion to apply force on the seal and thereby move the seal from the closed state to the open state and when the electric current is applied in the second current direction the electromagnet magnetically pulls the rod portion to remove the force on the seal and thereby move the seal from the open state to the closed state.

[0010] In a further embodiment of any of the foregoing embodiments, the force causes a portion of the seal to deflect and lift off of the seal seat so as to open the flow through the flow path.

[0011] A further embodiment of any of the foregoing embodiments includes controller connected with the electromagnet and configured to selectively apply the electric current in the first direction or the second direction.

[0012] In a further embodiment of any of the foregoing embodiments, the controller is configured to apply the electric current in the second current direction for a preset amount of time to move the seal from the open state to the closed state.

[0013] A valve according to an example of the present disclosure includes a valve body that has a

flow path, and a seal disposed in the flow path. The seal has a default closed state sealed against a seal seat so as to block flow through the flow path and an open state in which the seal permits flow through the flow path. An armature is disposed in the valve body. An electromagnet is configured to magnetically push the armature to apply force on the seal and thereby move the seal from the closed state to the open state. The force causes a portion of the seal to deflect and thereby separate from the seal seat so as to open the flow through the flow path.

[0014] In a further embodiment of any of the foregoing embodiments, the electromagnet includes first and second coil windings that are oppositely wound, and the armature includes a rod portion, a yoke comprised of a permanent magnet and a keeper that surrounds the permanent magnet, and a shell that encases the yoke.

[0015] In a further embodiment of any of the foregoing embodiments, the rod portion has a distal tip that is in contact with the seal and the seal is an elastomer.

[0016] In a further embodiment of any of the foregoing embodiments, the shell is non-ferromagnetic and is formed of a titanium-based alloy.

[0017] In a further embodiment of any of the foregoing embodiments, the shell is a material selected from the group consisting of titanium-based alloy, aluminum-based alloy, and polymer.

[0018] In a further embodiment of any of the foregoing embodiments, the electromagnet is configured to selectively receive electric current in a first current direction or a second, opposite current direction, wherein when the electric current is in the first current direction the electromagnet magnetically pushes the armature to apply the force on the seal and when the electric current is in the second current direction the electromagnet magnetically pulls the armature to reduce the force on the seal. The reduction in force causes the portion of the seal to move and thereby seat on the seal seat so as to close the flow through the flow path.

[0019] A rocket motor according to an example of the present disclosure includes a propellant tank holding propellant, a combustor, a nozzle attached with the combustor, a supply line that fluidly connects the propellant tank and the combustor, and a valve according to any of the foregoing embodiments situated in the supply line.

[0020] The present disclosure may include any one or more of the individual features disclosed above and/or below alone or in any combination thereof.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] The various features and advantages of the present disclosure will become apparent to those skilled in the art from the following detailed description. The drawings that accompany the detailed description can be briefly described as follows.

[0022] FIG. 1 illustrates an example of a rocket motor.

[0023] FIG. 2 illustrates a valve of the rocket motor in a closed state.

[0024] FIG. 3 illustrates a valve of the rocket motor in an open state.

[0025] FIG. 4 illustrates a valve with magnetic field lines.

[0026] FIG. 5 illustrates a region of the seal of the valve in a closed state.

[0027] FIG. 6 illustrates a region of the seal of the valve in an open state.

[0028] FIG. 7 illustrates a sectioned view through the seal of the valve.

[0029] FIG. 8 illustrates a valve in a latched state.

DETAILED DESCRIPTION

[0030] FIG. 1 illustrates a rocket motor 10 to demonstrate example aspects of a valve that is disclosed herein. It is to be appreciated, however, that applications other than a rocket motor will also benefit from this disclosure.

[0031] The rocket motor 10 generally includes a rocket motor body 12, a combustor 14, a nozzle

16 attached with the combustor **14**, a propellant tank **18** holding a propellant **18a**, and a supply line **20** that connects propellant tank **18** with the combustor **14**. There is a valve **22** in the supply line **20** and a controller **24** in communication with the valve **22** to control the operation thereof, namely control of the flow of the propellant **18a** to the combustor **14**. As will be appreciated, the rocket motor **10** will contain additional componentry that is beyond the scope of this disclosure.

[0032] FIG. 2 illustrates a sectioned view of an example of the valve **22**. The valve **22** includes a valve body **26** that has an inlet **26a**, an outlet **26b** and a flow path P that connects the inlet **26a** to the outlet **26b**. The valve body **26** may be formed of a single, unitary piece or multiple pieces that are secured to one another so as to bound the flow path P or a portion thereof.

[0033] The valve **22** further includes a seal **28** that is disposed in an outlet cavity **28a** along a portion of the flow path P. For example, the seal **28** is an elastomer that maintains good properties in the fluid that is to be conveyed through the valve **22**. In a rocket motor that utilizes hydrazine as the propellant **18a**, the elastomer may be ethylene propylene terpolymer. One example is known under the tradename AF-E-411 (Parker Hannifin), but other elastomers may be used as well. In FIG. 2, the seal **28** is in a default closed state in which it is sealed against a seal seat **30** so as to block flow through the flow path P. When the seal **28** is separated from the seal seat **30** (FIG. 3), flow is permitted through the flow path P.

[0034] The valve **22** also includes an armature **32** and an electromagnet **34**. The armature **32** is disposed in the valve body **26** and is moveable with respect to the seal **28**. The electromagnet **34** is connected to the controller **24** and includes first and second coil windings **34a/34b** that are oppositely wound. The coil windings **34a/34b** are connected in series. For example, there is a single continuous wire that has a portion that is wound in one direction to form the first coil winding **34a** and another portion that is wound in the opposite direction to form the second coil winding **34b**. The controller **24** may include hardware, software, or both, that are configured and/or programmed to carry out the functions herein with respect to control of the valve **22**. Hardware may include, but is not limited to, a micro-processor module, circuitry, control logic, memory module, and/or a power source.

[0035] The armature **32** is made up of several sub-components, including a rod portion **36**, a yoke **38**, and a shell **40** that encases the yoke **38** and prevents the fluid in the valve **22** from contacting the yoke **38**. The yoke **38** is comprised of a permanent magnet **42** and a keeper **44** that surrounds the permanent magnet **42**. The keeper **44** is made of a magnetic material. The rod portion **36** has a tip **36a** that is in contact with the seal **28**, the function of which will be described in further detail below.

[0036] The fluid that is intended to be conveyed through the valve **22** may not be compatible with the materials used in the yoke **38**. For instance, the fluid may accelerate corrosion of the material. As a non-limiting example in the rocket motor **10**, if hydrazine is to be used as the propellant **18a**, it may be desirable to avoid contact with stainless steel (e.g., the keeper **44**) and magnet. In this regard, the shell **40** hermetically encases the yoke **38** and thereby prevents the fluid from coming into contact with the sub-components of the yoke **38**. In one example, the shell **40** is a cladding that is applied around the yoke **38**. In other examples, the shell **40** may be a configuration of pre-fabricated pieces that are bonded to one another to encase the yoke **38**.

[0037] The shell **40** may be formed of a non-ferromagnetic material (not attracted to magnets) so as to avoid substantial interference with the magnetic fields of the permanent magnet **42** and the electromagnet **34**. For example, the material is selected from titanium-based alloy, aluminum-based alloy, or polymer. Other non-ferromagnetic alloys may also be used, such as non-magnetic nickel alloys (e.g., Inconel 625).

[0038] The material selected for the shell **40** is compatible with the fluid that is to be conveyed through the valve **22**, at least by comparison to one or more materials of the sub-components of the yoke **38**. Additionally, with the benefit of this disclosure one of ordinary skill in the art will recognize that the selected material will also have other strength and durability characteristics to

meet the design requirements of the particular implementation. For instance, aluminum alloy has good strength, durability, and corrosion resistance, and titanium alloy has excellent strength, durability, and corrosion resistance. Polymer may also be used for ease of manufacturing and chemical resistance but is generally not as strong as titanium or aluminum alloys.

[0039] In the operation of the valve **22**, the electromagnet **34** selectively receives an electric current in a first current direction or a second, opposite current direction. As indicated above, the controller **24** serves to operate the valve **22** and thus control the direction and magnitude of electric current. When the electric current is applied in the first current direction, the electromagnet **34** magnetically actuates the armature **32** to move the seal **28** from the default closed state (FIG. **1**) to the open state (FIG. **2**). When the electric current is applied in the second current direction the electromagnet **34** magnetically actuates the armature **32** to move the seal **28** from the open state to the closed state. As will be appreciated, the closed state is the default ready position in the illustrated examples, but a different default position could be used, such as a default ready open position or latched inactive condition.

[0040] FIG. **4** illustrates a sectioned view of the valve **22** with magnetic field lines **L1** that represent the magnetic fields generated by permanent magnet **42**. The electrical fields generated by the coil portions **34a/34b** of the electromagnet **34** cross the magnetic field of the permanent magnet **42**. By this interaction, the polarity, strength, and timing of the electrical fields generated by the electromagnet **34** operate to manipulate the magnetic field of the permanent magnet **42** to open and close the seal **28**.

[0041] FIGS. **5** and **6** demonstrate operation of the seal **28**. As shown in FIG. **5**, the seal **28** is disposed in the seal cavity **28a**. The seal **28** is trapped in the cavity **28a** in a state of compression so as to substantially prevent translation of the seal **28** along the axial direction of the rod portion **36** of the armature **32** and provide a pre-load against the seal seat **30**. In the closed state as in FIG. **5**, a seal surface **28b** bears against the seal seat **30**, thereby preventing flow from passing. The tip **36a** of the rod portion **36** may be in nominal contact with the seal **28** or spaced a short distance from the seal **28**. There is little or no force applied by the rod portion **36** to the seal **28**.

[0042] To open the seal **28**, the controller **24** applies electric current to the electromagnet **34** in the first current direction. The electrical field of the first coil portion **34a** interacts with the magnetic field of the permanent magnet **42** to magnetically push the armature **32** and thereby apply force on the seal **28** via the tip **36a** of the rod portion **36**. As shown in FIG. **6**, the force applied to the seal **28** causes a portion of the seal **28** to deflect and thereby lift the seal surface **28b** off of the seal seat **30**. With the seal surface **28b** lifted, fluid can then flow between the seal surface **28b** and the seal seat **30**, through flow channels **28c** (FIG. **7**) around the seal **28**, and then out of the outlet **26b** of the valve **22**. The terms “push” and “pull” as used herein refer to the direction of applied force and movement of the armature **32**, particularly the tip **36a** of the rod portion **36**, with respect to the electromagnet **34**. A push tends to move the tip **36a** away from the electromagnet **34** or in a direction out of the coil of the electromagnet **34**, while a pull tends to move the tip **36a** closer to the electromagnet **34** or in a direction into the coil of the electromagnet **34**.

[0043] The amount of force applied can be modulated via the amount of electric current and selection of cross-sectional area of the tip **36a** of the rod portion **36**. The amount of deflection for flow by the seal **28** is a function of the amount of force applied and the properties of the material of the seal **28**. The deflection provides a cross-sectional area for fluid flow by the seal **28**. In some examples, a metering function is provided at a smaller cross-sectional area of the flow path at a location upstream of the seal **28**, such as between the rod portion **36** and the walls of the orifice **46** (FIG. **5**). Thus, the metering is established between metal components and does not change even if the cross-sectional area for flow at the seal **28** were to vary due to changes in the material of the seal **28** over time, which may in turn permit use of a wider range of materials for the seal **28**.

[0044] To close the seal **28**, the controller **24** applies electric current to the electromagnet **34** in the second current direction, i.e., reverses polarity. The electrical fields of the coil portions **34a/34b**

interact with the magnetic field of the permanent magnet **42** to pull the armature **32** and thereby remove the force that was applied on the seal **28**. With the reduction in force, the portion of the seal **28** that had deflected to open the seal **28** then moves to thereby return to the closed state in which the seal interface **28b** seats against the seal seat **30** so as to block fluid flow.

[0045] The valve **22** operates to open and close without the use of a spring. In valves that have a solenoid coil and a spring, the force is proportional to the square of the coil current, and regardless of the electric current direction the armature is attracted to move in only one direction-against the spring. In the valve **22**, however, the force is linearly proportional to the electric current and the reversing of the electric current moves the armature **32** in opposite directions (push-pull). The push-pull ability eliminates the need for a spring, while the linear proportionality enables better control of the forces and thus enhanced valve performance.

[0046] In one example control scheme, the controller **24** executes an open command in which it applies a preset magnitude of electric current to the electromagnet **34** in the first current direction to open the seal **28**. The electric current is held constant for a period of time to maintain the seal **28** in the open state and thereby provide fluid flow (e.g., to the combustor **14** to generate thrust). When fluid flow is no longer desired, the controller **24** executes a close command in which it applies a preset magnitude of electric current to the electromagnet **34** in the second current direction to close the seal **28**. The electric current in the second direction is applied for a preset time period to remove the force from the seal **28**. At the end of the preset time period, the electric current is turned off (zero current). For example, the preset time period is 10-100 milliseconds, after which the magnetic field of the permanent magnet **42** maintains the position of the armature **32** such that the seal **28** is in the closed state.

[0047] In a further example, the valve **22** also has a third state in which the armature **32** is latched, i.e. a latched state. In the latched state, the armature **32** is moved to its fully upstream position against the inlet side of the valve body **26**. The permanent magnet **42** maintains the armature in this position with zero electric current in the electromagnet **34**. At this location, the tip **36a** of the rod portion **36** is substantially spaced apart from the seal **28**. For instance, prior to operational opening and closing of the valve **22**, the valve **22** may be subjected to vibrations that can cause chatter of the armature **32**. The chatter can cause the tip **36a** of the rod portion **36** to translate back-and-forth by a small distance. If the tip **36a** is in contact with the seal **28** or is in close proximity to the seal **28**, the chatter may cause the rod portion **36** to exert a transitory force on the seal **28**. This may cause the seal **28** to momentarily open and permit a small amount of fluid to leak by. However, in the latched state, the armature **32**, and thus the tip **36a**, are spaced from the seal **28** by a distance that is substantially greater than the amplitude of the chatter vibration. As a result, even if there is chatter of the armature **32**, it will not be able to contact the seal **28** or cause transitory leakage. Moreover, as the armature **32** is against the opening of the inlet **26a**, the armature **32** can provide a sealing function to limit inflow of the fluid into the valve **22**. The latched state is obtained by applying current in the second direction until the armature **32** contacts the inlet end of the body.

[0048] Although a combination of features is shown in the illustrated examples, not all of them need to be combined to realize the benefits of various embodiments of this disclosure. In other words, a system designed according to an embodiment of this disclosure will not necessarily include all of the features shown in any one of the Figures or all of the portions schematically shown in the Figures. Moreover, selected features of one example embodiment may be combined with selected features of other example embodiments.

[0049] The preceding description is exemplary rather than limiting in nature. Variations and modifications to the disclosed examples may become apparent to those skilled in the art that do not necessarily depart from this disclosure. The scope of legal protection given to this disclosure can only be determined by studying the following claims.

Claims

- 1.** A valve comprising: a valve body having a flow path; a seal disposed in the flow path, the seal having a default closed state sealed against a seal seat so as to block flow through the flow path and an open state separated from the seal seat so as to permit flow through the flow path; an armature disposed in the valve body and moveable with respect to the seal; and an electromagnet that is configured to selectively receive an electric current in a first current direction or a second, opposite current direction, wherein when the electric current is in the first current direction the electromagnet actuates the armature to move the seal from the closed state to the open state and when the electric current is in the second current direction the electromagnet actuates the armature to move the seal from the open state to the closed state.
- 2.** The valve as recited in claim 1, wherein the electromagnet includes first and second coil windings that are oppositely wound.
- 3.** The valve as recited in claim 1, wherein the armature includes a rod portion, a yoke comprised of a permanent magnet and a keeper that surrounds the permanent magnet, and a shell that encases the yoke.
- 4.** The valve as recited in claim 3, wherein the rod portion has a tip that is in contact with the seal.
- 5.** The valve as recited in claim 3, wherein the shell is non-ferromagnetic and is formed of a titanium-based alloy.
- 6.** The valve as recited in claim 5 wherein the seal is an elastomer.
- 7.** The valve as recited in claim 3, wherein the shell is a material selected from the group consisting of titanium-based alloy, aluminum-based alloy, and polymer.
- 8.** The valve as recited in claim 3, wherein when the electric current is in the first current direction the electromagnet magnetically pushes the rod portion to apply force on the seal and thereby move the seal from the closed state to the open state and when the electric current is applied in the second current direction the electromagnet magnetically pulls the rod portion to remove the force on the seal and thereby move the seal from the open state to the closed state.
- 9.** The valve as recited in claim 8, wherein the force causes a portion of the seal to deflect and lift off of the seal seat so as to open the flow through the flow path.
- 10.** The valve as recited in claim 1, further comprising a controller connected with the electromagnet and configured to selectively apply the electric current in the first direction or the second direction.
- 11.** The valve as recited in claim 10, wherein the controller is configured to apply the electric current in the second current direction for a preset amount of time to move the seal from the open state to the closed state.
- 12.** A valve comprising: a valve body having a flow path; a seal disposed in the flow path, the seal having a default closed state sealed against a seal seat so as to block flow through the flow path and an open state in which the seal permits flow through the flow path; an armature disposed in the valve body; and an electromagnet configured to magnetically push the armature to apply force on the seal and thereby move the seal from the closed state to the open state, the force causing a portion of the seal to deflect and thereby separate from the seal seat so as to open the flow through the flow path.
- 13.** The valve as recited in claim 12, wherein the electromagnet includes first and second coil windings that are oppositely wound, and the armature includes a rod portion, a yoke comprised of a permanent magnet and a keeper that surrounds the permanent magnet, and a shell that encases the yoke.
- 14.** The valve as recited in claim 13, wherein the rod portion has a distal tip that is in contact with the seal and the seal is an elastomer.
- 15.** The valve as recited in claim 13, wherein the shell is non-ferromagnetic and is formed of a

titanium-based alloy.

16. The valve as recited in claim 13, wherein the shell is a material selected from the group consisting of titanium-based alloy, aluminum-based alloy, and polymer.

17. The valve as recited in claim 13, wherein the electromagnet is configured to selectively receive electric current in a first current direction or a second, opposite current direction, wherein when the electric current is in the first current direction the electromagnet magnetically pushes the armature to apply the force on the seal and when the electric current is in the second current direction the electromagnet magnetically pulls the armature to reduce the force on the seal, the reduction in force causing the portion of the seal to move and thereby seat on the seal seat so as to close the flow through the flow path.

18. A rocket motor comprising: a propellant tank holding propellant; a combustor; a nozzle attached with the combustor; a supply line fluidly connecting the propellant tank and the combustor; and a valve situated in the supply line, the valve including: a valve body having a flow path connecting the inlet and the outlet, a seal disposed in the flow path, the seal having a default closed state sealed against a seal seat so as to block flow through the flow path and an open state moved from the seal seat so as to permit flow through the flow path, an armature disposed in the valve body and moveable with respect to the seal, and an electromagnet that is configured to selectively receive an electric current in a first current direction or a second, opposite current direction, wherein when the electric current is in the first current direction the electromagnet actuates the armature to move the seal from the closed state to the open state and when the electric current is in the second current direction the electromagnet actuates the armature to move the seal from the open state to the closed state.

19. The rocket motor as recited in claim 18, wherein the electromagnet includes first and second coil windings that are oppositely wound, the armature includes a rod portion, a yoke comprised of a permanent magnet and a keeper that surrounds the permanent magnet, and a shell that encases the yoke, and the shell is non-ferromagnetic and is formed of a titanium-based alloy.
